PSAI_105

TxSim: Transaction Simulator and Risk Detector for Polkadot

Team Name: Build Union
Payment Details:

- **DOT:** 13VM8zijwQcCSSTMuwdBEaM1mrqMz4xpDj2MkixUKBFmJxpm
- **USDC:** 13VM8zijwQcCSSTMuwdBEaM1mrqMz4xpDj2MkixUKBFmJxpm
- **Grant Level:** 2

Background

Build Union refined an earlier abandoned by HoodClock into **TxSim**, a transaction simulator and risk detector for Substrate WASM smart contracts and asset transactions in the Polkadot ecosystem. TxSim empowers Polkadot wallet users to preview transactions, identify risks, and optimize fees before signing, addressing a critical need for secure interactions with Polkadot's native contracts and assets.

The Problem

Polkadot's parachains (e.g., Moonbeam, Astar, Asset Hub) fuel DeFi and asset growth, but users face significant risks:

- **Blind Signing:** Transactions lack outcome visibility, causing extrinsic failures, reverts, or trapped assets.
- **Scams:** In 2024, phishing and pig butchering scams cost crypto users $3.6B, primarily on Ethereum, with Polkadot users facing similar threats from phishing and honeypot contracts that trap funds in fake liquidity or deceptive dApps.
- **Fee Waste:** Failed transactions incur fees in Polkadot's weight-based model, with excessive fees often signaling scams.

Existing tools fall short:

- **EVM-Centric:** Tenderly targets Ethereum, not Substrate.
- **Limited:** HoneyPot.is detects scam tokens post-loss; InstaDapp focuses on DeFi strategies.
- **Poor UX:** Developer-centric tools lack wallet integration.

TxSim addresses these gaps with real-time transaction previews, risk alerts for phishing and honeypots, and seamless Polkadot wallet integration.

TxSim Solution

TxSim is a user-friendly tool for Polkadot's Substrate-based smart contracts and asset transactions, offering:

- **Transaction Simulation:** Previews outcomes by simulating contract calls on Moonbeam and Astar parachains and extrinsic/runtime calls on Asset Hub (PVM), using `eth_call` for EVM-compatible chains (e.g., Moonbeam) and Substrate runtime queries for WASM/PVM chains (e.g., Astar, Asset Hub).

- **Risk Detection:** Flags extrinsic failures, reverts, and suspicious patterns (e.g., excessive fees, weight overflows, invalid parameters) across Polkadot parachains. *While TxSim's initial risk detection engine focuses on heuristic rules and WASM bytecode analysis, we plan to enhance detection capabilities over time by incorporating community reporting, machine learning models, and integration with external threat intelligence feeds. This phased approach will enable TxSim to adapt to emerging scam tactics and improve accuracy in identifying complex or zero-day threats.*

- **Fee Estimation:** Provides accurate fee estimates using Polkadot's weight-based model, with optimization suggestions.

- **Wallet Integration:** - Designed so wallet developers (e.g., Talisman, SubWallet) can easily connect to TxSim via the Polkadot.js API, without requiring separate accounts or additional

services.

- **Polkadot Focus:** Targets Moonbeam, Astar, and Asset Hub parachains, with modular SDK/APIs for dApp/wallet integration.

Ecosystem Fit

Polkadot's high transaction velocity and DeFi growth amplify scam and fee risks. TxSim provides Polkadot-native transaction safety, akin to Token Sniffer's scam detection for Solana, safeguarding users from honeypots and failed calls while fostering trust in parachains like Moonbeam, Astar, and Asset Hub. TxSim:

- Reduces failed transactions and scam losses.
- Enhances trust in dApps and wallets, improving L2 UX.
- Supports safe onboarding for millions of users.
- Delivers open-source infrastructure as a Polkadot public good.

Target Users

- **DeFi Users:** Preview staking, swapping, or minting transactions.
- **Retail Traders:** Avoid fee-wasting failures and honeypot scams.
- **Developers:** Test transactions during dApp development.
- **Wallet Platforms:** Integrate TxSim for enhanced user protection.

Technical Stack

- **Frontend:** Next.js + React, optimized for Polkadot wallet UX.
- **Backend:** Node.js + Polkadot.js API, with a simulation engine using RPC optimization and Ethers.js for Moonbeam's EVM compatibility.
- **Detection Engine:** Custom WASM bytecode parsing for Substrate contracts.
- **Database:** Scalable storage for over 100,000 contract signatures across Polkadot parachains.

Planned PVM (RISC-V) Support: While TxSim's initial MVP will focus on EVM (Moonbeam) and WASM (Astar), we've designed the architecture to support future integration of Asset Hub's upcoming RISC-V–based PolkaVM. Unlike WASM or EVM, PVM introduces new challenges — including limited tooling, evolving RPC standards, and the need for custom simulation emulators. However, our modular simulator engine is meant to plug in additional VMs, and we anticipate integrating PVM as the ecosystem matures, without requiring additional funding. This phased approach ensures TxSim stays aligned with the Polkadot Hub roadmap and remains compatible with its evolving execution environments.

Competitive Advantage

TxSim uniquely combines transaction simulation, risk detection, and user-friendly UX for Polkadot's Substrate ecosystem. The table below compares TxSim to competitors:

Feature	TxSim	Tenderly	HoneyPot.is	InstaDapp
Substrate Contract Support	✅	❌	❌	❌
Transaction Simulation	✅	✅ (EVM only)	❌	❌
Real-Time Risk Detection	✅	❌	✅ (Tokens only)	❌
Wallet Integration	✅	❌	❌	✅
Polkadot Parachain Support	✅	❌	❌	❌

Why TxSim?

TxSim eliminates blind transactions in Polkadot, reducing scams and fee waste while enhancing user trust. By supporting Moonbeam, Astar, and Asset Hub, TxSim strengthens Polkadot's security-first UX, enabling safer DeFi and asset interactions. With a live demo (TxSim Demo) and open-source approach, TxSim is poised to set a new standard for transaction safety in Polkadot.

Team

Build Union is a collaborative collective of talented developers, designers, strategists, and innovators dedicated to shaping the future of the decentralized web. As an aggregate team of Web3 builders, Build

Union brings together diverse expertise to create cutting-edge blockchain applications, DAO services, and decentralized solutions, focused on fostering innovation and accelerating Web3 adoption.

Members

Emmanuel — Project Lead
Role: Project manager and blockchain architect with deep experience in DAO tooling and transparency infrastructure. Combines hands-on engineering with leadership in dApp development and open governance tooling.
LinkedIn: linkedin.com/in/emmanuel-eclipse
Contact: virtualstrikingly@gmail.com

Golden K. — Machine Learning Developer
Role: Develops Python-based ML pipelines for anomaly detection and data analysis using pandas, NumPy, SVM, and PyTorch.
GitHub: github.com/golden-k22

Eliud Munyala — Senior Data Engineer & ML Specialist
Role: Leads data engineering at SpaceAI, specializing in data ingestion, transformation, and validation.
GitHub: github.com/meaLuda
LinkedIn: linkedin.com/in/eliud-munyala

Marko — UI/UX Designer
Role: Designed over 1,000 websites and data dashboards, leading UX design for WatchDog.
Portfolio: hipfolio.co
Dashboards: hipfolio.co/dashboard.html

Awais Nazir — Full-Stack Developer (dApps)
Role: Specializes in smart contracts, React, Next.js, and Node.js for decentralized applications.
LinkedIn: linkedin.com/in/awais-nazir

WayToTheVenus — Smart Contract Developer
Role: Develops modular, audit-ready smart contracts.
GitHub: github.com/waytothevenus

CryptoStar7 — Blockchain & Full-Stack Developer
Role: Builds backend blockchain logic, frontend application layers, smart contract architecture, and full-stack Web3 tooling.

Legal Structure

- **Registered Address:** Flat 1 Park View, 693 Altrincham Road, Manchester, England, M23 9AA
- **Registered Legal Entity:** Build Union LTD **Team Code Repositories**
  Technical and architectural planning is ongoing, with public code repositories to be established upon Milestone 1 completion.

Budget

Overview

- **Total Estimated Duration:** 3 months
- **Full-Time Equivalent (FTE):** 1 (2 developers part-time, ~20 hours/week each per milestone)
- **Total Costs:** 28,000 USD
- **DOT %:** 50% (14,000 USD equivalent in vested DOT, 14,000 USD in USDC)

Milestone 1 — Core Simulation Engine

- **Estimated Duration:** 1 month
- **FTE:** 1
- **Costs:** 10,000 USD

Deliverables

Number	Deliverable	Specification

0a.	License	MIT
0b.	Documentation	We will provide both inline documentation and a user guide explaining how to run the simulation engine, estimate fees, and interpret outputs.
0c.	Testing and Guide	Core functions will be covered by unit tests (≥80% coverage). We will include a guide to run and validate them.
0d.	Docker	A Dockerfile will be provided for quick testing and deployment of the simulation engine.
1.	Simulation Engine	Initial transaction simulation engine targeting Moonbeam (EVM) and Astar (WASM) networks, with an architecture designed for future compatibility with Substrate pallets and runtime calls (e.g., Asset Hub PVM).
2.	Fee Estimation Module	First implementation of the fee estimation logic based on simulated outcomes.

Milestone 2 — Fee Optimization and Testing

- **Estimated Duration:** 1 month
- **FTE:** 1
- **Costs:** 10,000 USD

Deliverables

Number	Deliverable	Specification
0a.	License	MIT
0b.	Documentation	Updated documentation to reflect new features and enhancements.
0c.	Testing and Guide	≥90% test coverage with an expanded test suite to handle edge cases.
0d.	Docker	Updated Dockerfile for the enhanced version of the simulation engine.
1.	Fee Optimization	Advanced logic to recommend optimal transaction parameters (e.g., gas limits, max fees) based on simulation data.
2.	Performance Tuning	Enhancements in simulation speed and accuracy for an improved development experience.

Milestone 3 — SDK and Integration APIs

- **Estimated Duration:** 1 month
- **FTE:** 1
- **Costs:** 8,000 USD

Deliverables

Number	Deliverable	Specification
0a.	License	MIT
0b.	Documentation	A complete SDK/API reference and a guide for integration into dApps or wallets.
0c.	Testing and Guide	Unit tests (≥80%) for the SDK and API modules with a usage walkthrough.
0d.	Docker	Dockerfile for testing the SDK via a minimal backend+frontend demo.
0e.	Article	We will publish a developer-focused article demonstrating the SDK integration in a real-world Moonbeam/Astar dApp.
1.	JavaScript SDK	Modular JS SDK with APIs for dApps/wallets to integrate TxSim’s simulation and fee estimation, supporting EVM, WASM smart contracts, and Substrate pallet extrinsics.
2.	Frontend Integration	Examplefrontend showcasing SDK integration with example transactions and simulation results.

Adoption Strategy

- **Wallet Partnerships:** Engage Talisman and SubWallet for integration during Milestone 2.
- **Beta Program:** Launch with 1,000 DeFi users and developers on Moonbeam, Astar, and Asset Hub post-Milestone 2, promoting the live demo (TxSim Demo) to showcase transaction simulation and risk detection.
- **Community Engagement:** Host AMAs on X, post on Polkassembly, and present at Polkadot events (e.g., Sub0).
- **Open-Source:** Release simulation engine and SDK on GitHub under MIT license.

Risks & Mitigation

Risk	Mitigation
WASM parsing complexity	Phased testing with known contract data
Wallet integration delays	Early engagement with wallet teams in M2
Asset Hub integration challenges	Leverage Polkadot.js for PVM compatibility

Conclusion

TxSim eliminates blind transactions in Polkadot, preventing scams and fee waste while boosting user trust. By supporting Moonbeam, Astar, and Asset Hub, TxSim strengthens Polkadot’s security-first UX, to make DOT transactions safe.**